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A biologist investigated the stimulation of a Pacinian corpuscle in the skin of a fingertip.
She used microelectrodes to measure the maximum membrane potential of a Pacinian corpuscle and its sensory neurone when different pressures were applied to the fingertip.

Figure 4 shows the Pacinian corpuscle, its sensory neurone and the position of the microelectrodes.

Figure 4

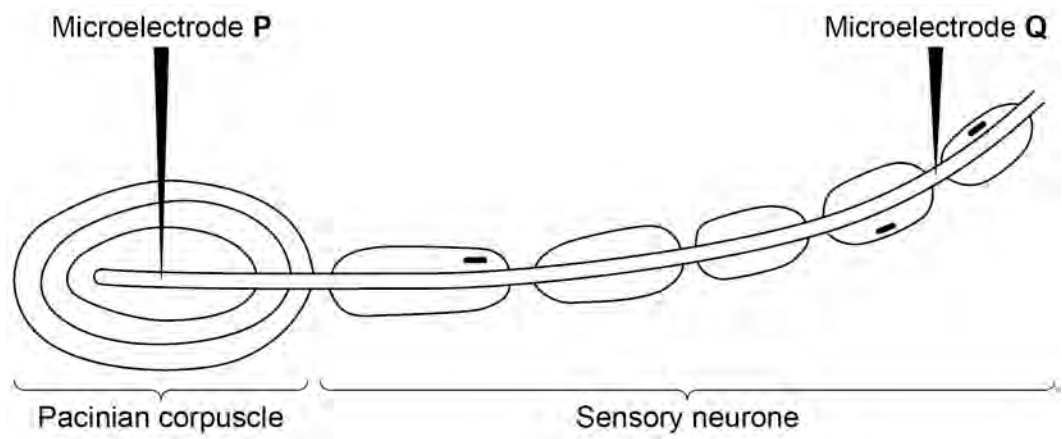


Table 2 shows some of the biologist's results.

Table 2

Pressure applied to the fingertip	Membrane potential at P / millivolts	Membrane potential at Q / millivolts
None	-70	-70
Light	-50	-70
Medium	+30	+40
Heavy	+40	+40

0 4 . 1 Explain how the resting potential of -70 mV is maintained in the sensory neurone when no pressure is applied.

[2 marks]

0	4	.	2
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Explain how applying pressure to the Pacinian corpuscle produces the changes in membrane potential recorded by microelectrode **P**.

[3 marks]

[Extra space]

0	4	.	3
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The membrane potential at **Q** was the same whether medium or heavy pressure was applied to the finger tip. Explain why.

[2 marks]

0	4	.	4
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Multiple sclerosis is a disease in which parts of the myelin sheaths surrounding neurones are destroyed. Explain how this results in slower responses to stimuli.

[2 marks]
